

LESSON 12.4

VOLUME OF RIGHT CIRCULAR CYLINDERS – PART 1



LESSON INTRODUCTION

MA.7.GR.2.3 Solve mathematical and real-world problems involving volume of right circular cylinders.

LEARNING TARGETS

- I can find the volume of a right circular cylinder.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.

ESSENTIAL QUESTIONS

- How are capacity and volume related?
- How is the area of the cross-section of a right circular cylinder found in the formula for volume of a right circular cylinder?

LESSON NARRATIVE

Students engage in an interactive activity with pennies (or similar manipulatives) to derive the formula for a right circular cylinder. Having derived the formula of the volume of the right circular cylinder, students will find the volume of the cylinder given the measure of its radius and height. Students will practice finding the volume of several cylinders.

COMPONENT SUMMARY

12.4.1 WARM-UP (~5 MIN.)

Comparison of measurements

12.4.3 EXPLORATION (10 MIN.)

Exploration of the volume of a circular cylinder

12.4.5 YOUR TURN! (10 MIN.)

Practice calculating the volume of a circular cylinder

12.4.2 ACTIVITY (10 MIN.)

Collaborative comparison of cylindrical shapes via penny stack

12.4.4 GUIDED PRACTICE (~10 MIN.)

Practice determining the parts that make up the volume of a circular cylinder

12.4.6 WRAP-UP (~5 MIN.)

Self-Assessment of understanding of learning target

RESOURCES

GLOSSARY

Key Terms:

- Capacity

Additional Terms:

- Volume of a cylinder

MATERIALS

- Pennies (or other circular objects such as Necco wafers)
- Calculators

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to make the transference from a stack of pennies/nickels to an empty cylinder.
- Students may struggle with the concept of the capacity in the circular base of the empty cylinder.

THINGS TO CONSIDER

- Ensure consistent use of the correct vocabulary associated with the standard.
- Consider your students' ability to connect the conceptual understanding of volume to the stacks of pennies. Students may not relate the volume formula $V=Bh$ to the volume formula $V=\pi r^2 h$ connecting that $B=\pi r^2$. Be purposeful in helping students build this connection. If necessary, use a whole-group demonstration starting with one penny and building up or doing the activity as a whole class.
- If time is an issue, then consider assigning 12.4.5 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

12.4.2 Activity positions students to notice volume of a cylinder as it relates to a stack of pennies. Consider applying the Notice and Wonder routines for each student group, positioning students to discover the relationship between the area of the circular cross-section and volume of different right circular cylinder.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.1.1 Participate in Effortful Learning

Students will be working to understand the formula for volume of a right circular cylinder as the product of the area of the base and the height. As students are exploring cylinders, they will need to ask clarifying questions to complete given activities. Encourage students to engage in routines that produce collective learning while recognizing and praising students who exhibit these qualities with positive narration. "I hear Nadia asking good questions to her partner. I see Karah drawing a model to show her partner."

DIFFERENTIATE INSTRUCTION

In 12.4.2 Activity and 12.4.3 Exploration students are provided with visual and kinesthetic entry-points as they find the volume of a cylinder represented by a stack of pennies. Provide cylindrical objects as visual aids to help students visualize area and volume by handling real-life examples of cylinders. Provide an auditory entry-point by reading problems aloud, modeling an example, and having students discuss their strategies and answers.

12.4.1 WARM-UP: WOULD YOU RATHER?

ENRICHMENT

Ask the following question to connect the Warm-Up to the lesson:

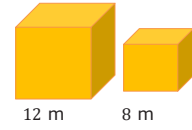
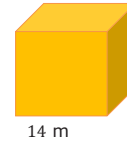
- To compare the weight of the gold, is it better to find the surface area or the volume?



12.4.1 Warm-Up: Would You Rather?

Gold is worth a significant amount of money. Its value changes with the market. For example, on January 28, 2021 the value of 1 ounce of gold was nearly \$2000.

- Would you rather have one cube made of solid gold measuring 14 meters (m) on each side, or two cubes made of solid gold, one measuring 12 m on each side and one measuring 8 m on each side?



Answers will vary.

- Explain your choice.

Answers will vary. I would take the 14-meter square because it is the biggest square. I would take the two 12 meter and 8-meter squares because they have more surface area.



12.4.2 Activity: Penny Stacks – Volume of a Right Circular Cylinder

1. Stack the given pennies to form a right circular cylinder.
 - a. How many pennies high is the stack?
Answers will vary depending on how many pennies they were given in this activity.

- b. Compare your stack of pennies with your partner's stack. Suppose both stacks are hollow such that liquid could be poured into them. Discuss with your partner which stack would hold more liquid. Write a summary of your discussion.



Answers will vary. The stack with more pennies or the higher stack will hold more liquid.



Capacity is the amount of space that can be filled in a container. Both capacity and volume are used to measure three-dimensional spaces.

Capacity usually refers to fluid measures, whereas volume is measured in cubic units.

- c. Complete the statement.
The capacity, or volume, of the empty cylinder depends on

- ☐ the radius of each coin.
- ☐ the height of the coin stack.
- ☒ both the height and radius of the coin stack.

- d. Circle the penny stack that has the greatest volume.



12.4.2 ACTIVITY: PENNY STACKS – VOLUME OF A RIGHT CIRCULAR CYLINDER

TEACHER MOVES

Notice and Wonder

Students should complete their penny stack and take time to individually write down observations about their stack and their partner's stack.

- What do you notice about the height of your stack and the height of your partner's stack?
- What do you wonder about the amount of liquid the cylinder could hold?

Facilitate a discussion where groups share their Notice and Wonder Observations.

TEACHER MOVES

Students may struggle to conclude that the volume will depend on the number of coins/circles which is the height and area of the base. Relate the height to how high the circles are stacked. Ask students what is changing as more pennies are being added.

QUESTIONING

Challenge students that complete the activity early to create a method or formula for determining the volume of a right circular cylinder. Students can compare their descriptions or formulas provided later in the lesson.

12.4.3 EXPLORATION: DISCOVERING THE VOLUME OF A RIGHT CIRCULAR CYLINDER

TEACHER MOVES

Exploration is intentionally constructed for discovery-based learning. Question prompts are intentionally scaffolded to orient students to content and each other while making sense of their discoveries through the lens of the formula for volume of a right circular cylinder.

After students have completed the activity, consider prompting students to share the following:

- One thing they learned
- One thing they still have a question about
- How they would describe the formula for the volume of a right circular cylinder.

DIFFERENTIATE INSTRUCTION

Consider allowing students to use their penny stack and an empty cylinder as visual aids. Cylinders can be made from paper, cardstock, or cardboard. The empty cylinders can be cut to reveal that the cross-section is a circle.

ENRICHMENT

Consider asking students the following questions:

- What is the capacity of a circle?
- How is the volume of a cylinder created?
- What dimension creates the volume of a cylinder?

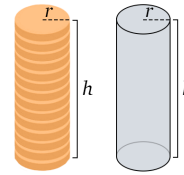
QUESTIONING

Consider a foldable, insert, and/or interactive notebook for students to collect information about volume of a cylinder, including recording formulas. Examples, drawings of cylinders and cross-sections, and verbal descriptions will provide multiple pathways for students to engage with the content in meaningful ways that encourage conceptual understanding over rote memorization.



12.4.3 Exploration: Discovering the Volume of a Right Circular Cylinder

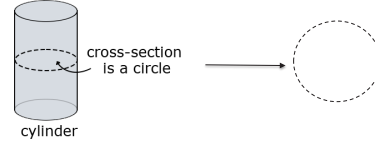
1. A penny stack with radius, r , and height, h , and an empty cylinder with the same dimensions are shown.



Complete the statement.

The volume of the penny stack is ☐ equal to ☐ not equal to the volume of the empty cylinder.

2. A cylinder can be sliced to create a cross section.



Complete the statements.

The cross section of the cylinder shown is the same size as the ☐ circular base ☐ lateral area ☐ height of the cylinder.

The expression ☐ πr^2 ☐ $2\pi r$ represents the area of the cross section of the cylinder.

3. With your partner, discuss the relationship between one of the cross sections of the cylinder and the capacity of the entire cylinder.

- a. Summarize your discussion.

Answers will vary. Each cross section represents the area of that circular piece. If we add all the circular pieces, we would get the volume of the cylinder.

- b. Complete the statements.

Area measures the two-dimensional space an object

takes up. The ☐ radius ☐ height of the empty cylinder

makes it a three-dimensional space, as multiple layers with the same area are stacked upon each other. To find the **volume of a cylinder**, multiply the **area** of the base by the **height** of the cylinder.

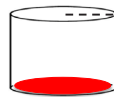
4. Which formula represents the volume of a right cylinder?
 - ☐ $V = r \cdot h$, where r is the radius and h is the height.
 - ☐ $V = Ch$, where C is the circumference of the base and h is the height.
 - ☐ $V = Bh$, where B is the area of the base and h is the height.



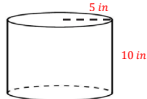
12.4.4 Guided Practice: Calculating the Volume of a Right Circular Cylinder

1. A diagram of a right circular cylinder is shown with the radius, r , of 5 meters (m.) and height, h , of 9 m.

$$h = 9$$



- Shade in the base of the cylinder.
 - Calculate the area of the base. Leave terms of π and use the appropriate units.
 $A = \pi r^2 = \pi 5^2 = 25\pi \text{ m}^2$
 - Calculate the volume of the cylinder. Leave in terms of π and use the appropriate units.
 $V = (25\pi)(9) = 225\pi \text{ m}^3$
2. Consider a cylinder with a radius 5 inches (in.) and a height of 10 in.
- Sketch the cylinder and label its dimensions.



- Calculate the volume of the cylinder using $\frac{22}{7}$ to approximate π . Round to the nearest thousandth and use appropriate units.

$$V = \pi r^2 h = \left(\frac{22}{7}\right)(5^2)(10) = \frac{5500}{7} \text{ in}^3 = 785.714 \text{ in}^3$$

12.4.4 GUIDED PRACTICE: CALCULATING THE VOLUME OF A RIGHT CIRCULAR CYLINDER

TEACHER MOVES

Question 1 is intentionally scaffolded to make the connections between the area of the base of a right circular cylinder and the volume. Students have not been formally introduced to the formula $V=Bh$. Through 12.4.3 Exploration and 12.4.4 Guided Practice, guide students to discover that volume is the area of the base times the height. Consider if a more formal introduction to the formulas will be helpful to your students.

DIFFERENTIATE INSTRUCTION

Consider placing students in groups and structuring this component as a jigsaw:

- Assign half of the groups question 1.
- Assign the other half question 2.
- Each group should record their answer on chart paper OR work can be displayed on an interactive monitor or similar device.
- Pick a few groups from each question to discuss their methods and solutions.

12.4.5 YOUR TURN!

TEACHER MOVES

Students, using the formulas given, should be able to independently determine the volume of each cylinder. Prompt students to notice the similarities of the horizontal and vertical cylinders. Specifically, that the radius is always associated with the area of the circle and height is the length of the cylinder.

DIFFERENTIATE INSTRUCTION

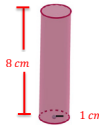
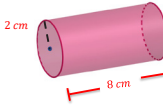
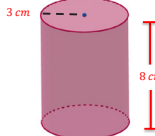
Students that completed the challenge work in 12.4.2 Activity can compare their equation(s) with the volume formula. Questions to pose:

- Was your description or equation similar to the formula used to find the volume of cylinders?
- If not, what portions did you miss or add?



12.4.5 Your Turn!

- Suppose three cylinders each have the same height of 8 centimeters (cm.). Cylinder 1 has a radius of 1 cm. Cylinder 2 has a radius of 2 cm. Cylinder 3 has a radius of 3 cm. Label the cylinders with their heights and radii, then find the volume of each cylinder. Use 3.14 to approximate π . Round your answers to the nearest tenth and include appropriate units.

Cylinder Number	Label with Height and Radius	Volume
1		$V = (3.14)(1^2)(8)$ $= 25.1 \text{ cm}^3$
2		$V = (3.14)(2^2)(8)$ $= 100.5 \text{ cm}^3$
3		$V = (3.14)(3^2)(8)$ $= 226.1 \text{ cm}^3$



12.4.6 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the Exploration and Guided Practice that I knew were ...	Something I wondered during the Exploration and Guided Practice was ...	Something I learned during the Exploration and Guided Practice was ...
Answers will vary.	Answers will vary.	Answers will vary.

12.4.6 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

- For students who might struggle representing their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson.
- Students could draw and label a cylinder, provide measurements for radius and height, and calculate volume.
 - Students could measure the radius and height of the pennies used in the penny stack and calculate the area.
 - Students can record a verbal explanation of the prompts if technology is available.
 - All forms can show an equal level of rigor and conceptual understanding.